

SEAN JAFRI

Robotics Hardware Engineer • Mechanical Design • Test Automation

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SUMMARY

Robotics hardware engineer with 8+ years designing and deploying robotic test fixtures, automated validation systems, and mechatronic prototypes for HP Inc.'s Commercial Systems group. End-to-end ownership from concept sketch through factory-floor deployment: SolidWorks CAD with GD&T, in-house MJF printing on HP Multi Jet Fusion 4200, vendor CNC, manual machining, electronics integration (Eagle PCB layout, Arduino, solenoids, servos, linear actuators). Track record of substantial cost reduction — 87% on flagship laptop test fixture, 67% on robot controller — without sacrificing reliability.

CORE SKILLS

Mechanical Design: SolidWorks (8+ yrs daily), GD&T per ASME Y14.5, DFM/DFA, mechanism & linkage design, tolerance stack-ups

Fabrication: HP Multi Jet Fusion 4200 operator, FDM & SLA prototyping, 80/20 extrusion, manual mill machining, vendor CNC

Mechatronics: Servo, stepper, and linear actuators; solenoids and coil drivers; sensor integration; Arduino firmware; PCB layout (Eagle)

Test & Validation: Specialized fixture design, automated V&V infrastructure, root-cause analysis, reliability growth, traceability to engineering requirements

PROFESSIONAL EXPERIENCE

Robotics Hardware Engineer (Contract via Mphasis) HP Inc., Houston, TX Apr 2018 — Present

- Lead mechanical design and deployment of **14+ robotic test fixtures** used daily by HP's Commercial PC Validation team to verify laptops, docks, and accessories at production scale.
- Reduced fixture cost from \$3,500 to \$450 per unit (**~87% reduction**) across four design generations; cut setup time from 10 min to 2 min and doubled deployment density per bench.
- Designed Eagle-routed shield over Arduino Mega 2560 with touchscreen UI to drive twelve actuators per controller, bringing per-controller cost from \$300 to \$100 while tripling channel count.
- Own end-to-end product cycle: SolidWorks CAD with GD&T, MJF/SLA/FDM prototyping (operate HP Multi Jet Fusion 4200), vendor CNC, in-house machining, electronics integration, and direct factory-floor deployment with test technicians.

SELECTED PROJECTS

- **Deep Purple Dock Test Fixture** — 12-actuator robotic dock-cycler exercising USB-C, USB-A, HDMI, and DisplayPort connectors on HP docking stations; reconfigurable for 8-, 9-, and 12-port variants; custom Eagle PCB.
- **Robotic Laptop Test Fixture (4 generations)** — Iterated from bespoke machined-aluminum prototype to current production fixture using off-the-shelf laptop holder with MJF brackets; \$3,500 → \$450 per unit, 87% cost reduction over 8 years.
- **Keyboard Bot** — Compact fixture driving 120+ solenoids to actuate every key on a laptop keyboard; enables automated typing-fatigue validation, key-rollover testing, and BIOS hotkey verification.
- **Universal Laptop Cradle** — Replaced \$350 model-specific cradles with a \$25 set of MJF-printed corner brackets that hold any laptop, eliminating per-model retooling across the validation fleet.
- **Mega II Robotic Test Controller** — Custom Arduino Mega shield + touchscreen UI controlling 12 actuators per controller; \$300 → \$100 unit cost replacing the previous 4-bot controller plus per-actuator bot boards.
- **Additional Hardware** — USB-C Flip Bot (planetary-gear plug rotator), Multiport Bot (3-port concurrent USB-A actuator array), Low Power Switch (4-channel modular USB enable/disable), Varda Reviver (USB-attached laptop revive device with Eagle PCB).

EDUCATION

B.S. Mechanical Engineering The University of Texas at Arlington

May 2017

LEADERSHIP & UNIVERSITY PROJECTS

- **President / VP / Team Lead, UTA Mars Rover Team** (ROVER Society, 2014–2017) — Led 40+ member team designing planetary rover for The Mars Society's University Rover Challenge (competed 2014, 2015, 2017); integrated mechanical, electrical, and control systems; raised \$19,000+ in corporate and academic sponsorships.
- **Senior Design** — Designed and prototyped UTA's first remotely-piloted Unmanned Vehicle System (UVS), a tracked ground robot with onboard sensor suite and wireless telemetry.